1 / 6

A wait needs a path to progress.

Three philosophers each need two exclusive forks.

> Chosen ownership and participation examples · not GPU scheduling or timing

Required forks

A	B	C
Needs 0 and 1	Needs 1 and 2	Needs 0 and 2

Lower number → higher number

Acquire both → eat once → release both

Chosen round-robin opportunities · not a hardware schedule

Copyright © 2026 Christopher Snow

2 / 6

A common order leaves a way forward.

B holds both forks and can eat at its next opportunity.

> Chosen ownership and participation examples · not GPU scheduling or timing

Exclusive ownership

Fork 0	Fork 1	Fork 2
Owner A	Owner B	Owner B

Philosopher state

A	B	C
Holds 0	Both held; can eat	Holds none

A → fork 1 → owner B

C → fork 0 → owner A

B can finish → release forks → unblock A

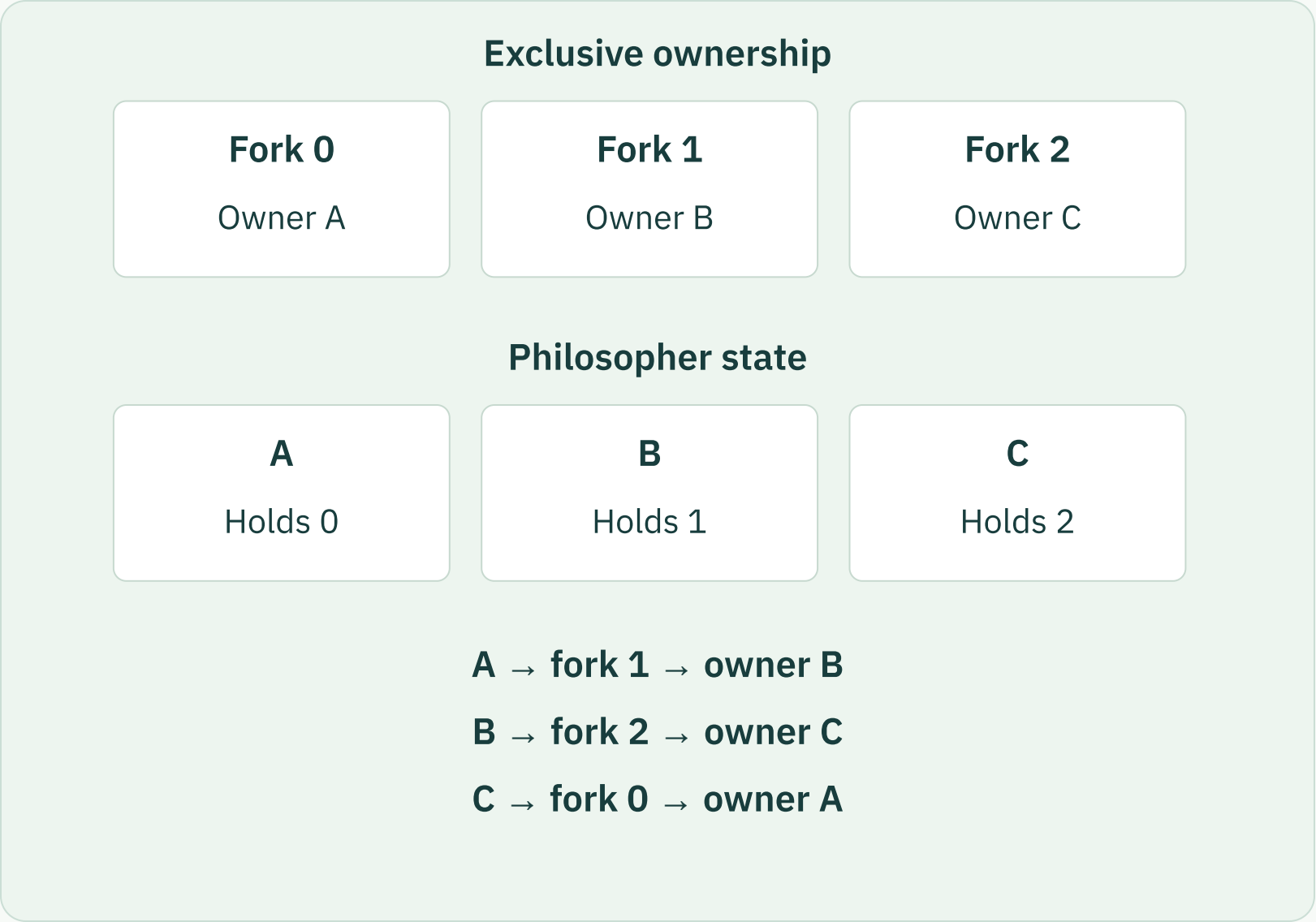
Copyright © 2026 Christopher Snow

3 / 6

A circular wait has no releaser.

Taking each philosopher’s own fork first closes the cycle.

> Chosen ownership and participation examples · not GPU scheduling or timing



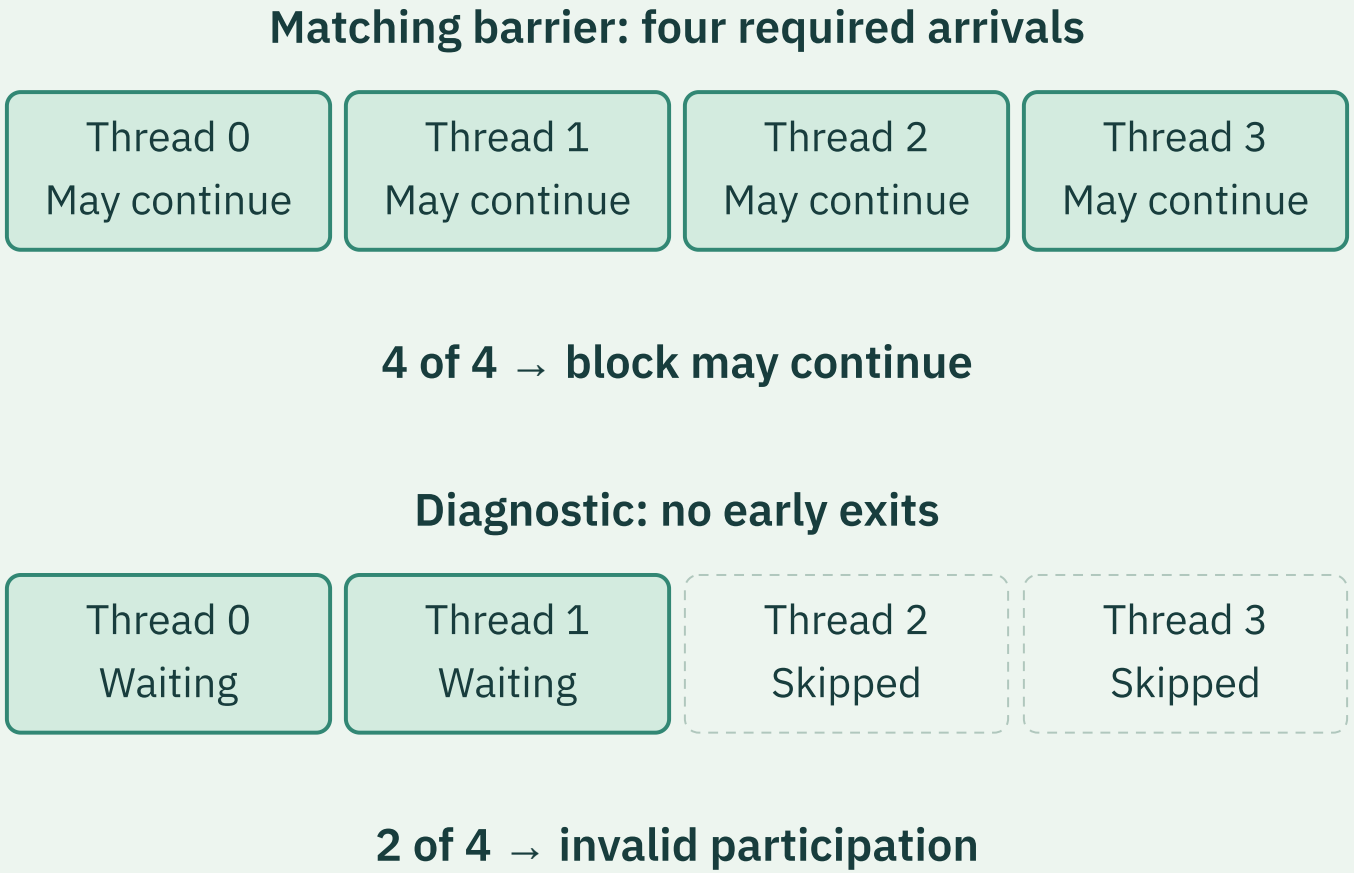
Copyright © 2026 Christopher Snow

4 / 6

A barrier requires matching participation.

All four threads arrive; skipped calls cannot supply arrivals.

> Chosen ownership and participation examples · not GPU scheduling or timing



Copyright © 2026 Christopher Snow

5 / 6

Pause: can the producer run?

One resident slot; A waits for a flag only B can set.

> Chosen ownership and participation examples · not GPU scheduling or timing

Chosen ordinary-block capacity

Resident A	Capacity	Pending B
Waits for B’s flag	1 block	Must set the flag

A needs B → B needs the occupied slot

Does resource order solve this wait?

Predict before the comparison

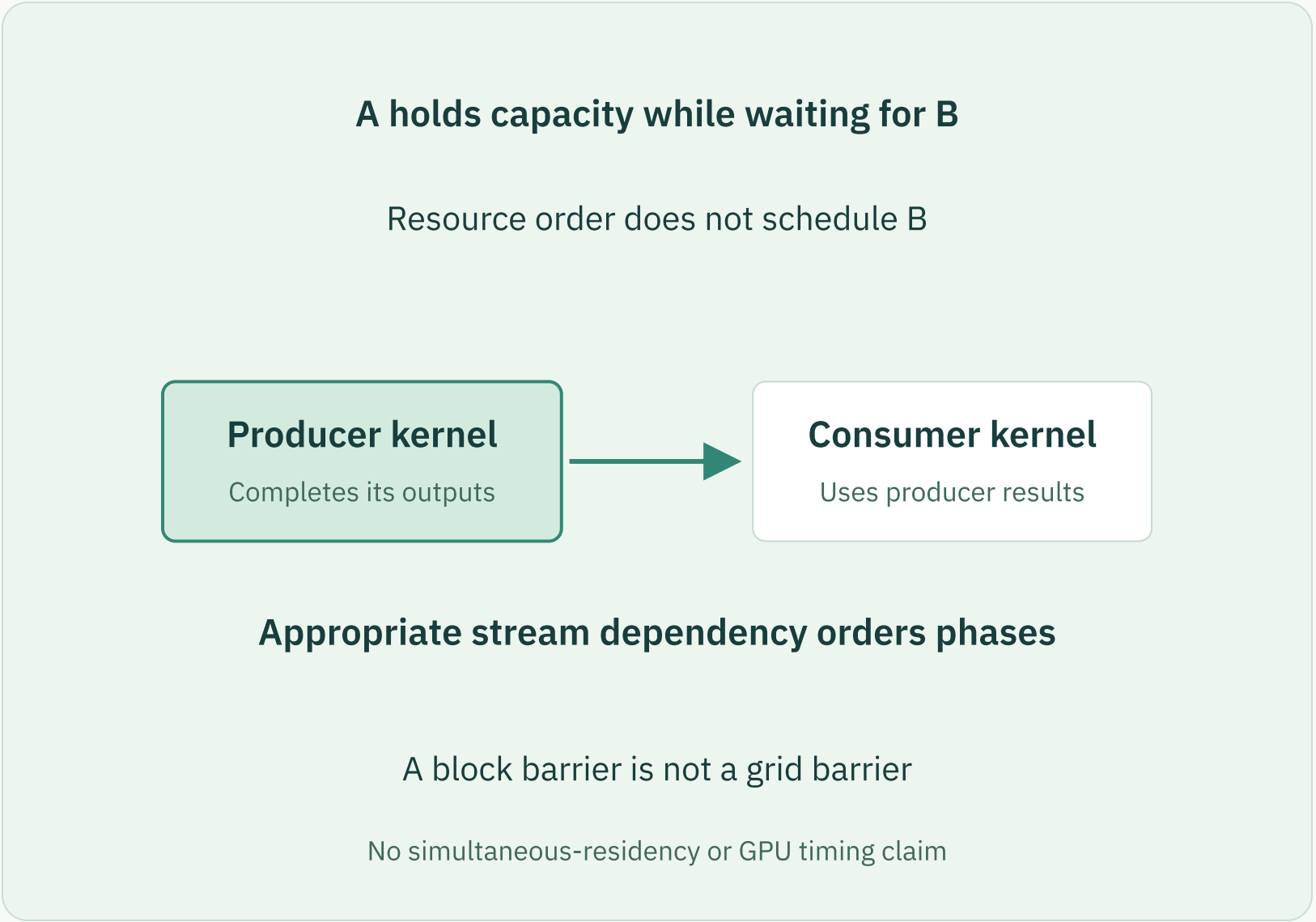
Copyright © 2026 Christopher Snow

6 / 6

Order phases without occupying their capacity.

Finish the producer phase before its consumer uses the result.

> Chosen ownership and participation examples · not GPU scheduling or timing



Copyright © 2026 Christopher Snow